

Multivariate Location Tests: Comparing Several Mean Vectors

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8. Two-Way MANOVA

Content adapted from:

Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed).

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Univariate Reminder: Student's Two-Sample t Tests

Remember: there are two types of two-sample t tests:

- **Dependent samples:** two repeated measures from same subject
- **Independent samples:** measurements from two different groups

For the dependent samples t test, we test

$$H_0 : \mu_d = \mu_0 \quad \text{versus} \quad H_1 : \mu_d \neq \mu_0$$

where $\mu_d = E(d_i)$ with $d_i = x_{i1} - x_{i2}$ denoting a difference score.

For the independent samples t test, we test

$$H_0 : \mu_x - \mu_y = \mu_0 \quad \text{versus} \quad H_1 : \mu_x - \mu_y \neq \mu_0$$

where $\mu_x = E(x_i)$ and $\mu_y = E(y_i)$ are the two population means.

Multivariate Extensions of Two-Sample t Tests

In this section, we will consider multivariate extensions of the (univariate) two-sample t tests.

Similar to the univariate case, the multivariate dependent samples T^2 test performs the one-sample test on a difference score.

The independent samples case involves a modification of the T^2 statistic, and can be extended to $K > 2$ samples.

- $K > 2$ is a multivariate analysis of variance (MANOVA) model

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Univariate Reminder: Dependent Samples t test

To test the hypotheses $H_0 : \mu_d = \mu_0$ versus $H_1 : \mu_d \neq \mu_0$ we use

$$t = \frac{\bar{d} - \mu_0}{s_d / \sqrt{n}}$$

where $\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$ and $s_d^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$ with $d_i = x_{i1} - x_{i2}$.

The $100(1 - \alpha)\%$ CI for the population mean of the difference score is

$$\bar{d} - t_{n-1}(\alpha/2) \frac{s_d}{\sqrt{n}} \leq \mu_d \leq \bar{d} + t_{n-1}(\alpha/2) \frac{s_d}{\sqrt{n}}$$

Multivariate Difference Scores

Let x_{ijk} denote the k -th repeated measurement of the j -th variable collected from the i -th subject, and define $\mathbf{x}_{ki} = (x_{i1k}, \dots, x_{ipk})^\top$.

The i -th subject's vector of difference scores is defined as

$$\mathbf{d}_i = \mathbf{x}_{1i} - \mathbf{x}_{2i} = \begin{pmatrix} x_{i11} - x_{i12} \\ x_{i21} - x_{i22} \\ \vdots \\ x_{ip1} - x_{ip2} \end{pmatrix}$$

and note that $\mathbf{d}_i \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_d, \boldsymbol{\Sigma}_d)$ assuming the subjects are independent.

Hotelling's T^2 for Difference Score Vectors

Given that $\mathbf{d}_i \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_d, \boldsymbol{\Sigma}_d)$, the T^2 statistic has the form

$$T_d^2 = n(\bar{\mathbf{d}} - \boldsymbol{\mu}_d)^\top \mathbf{S}_d^{-1}(\bar{\mathbf{d}} - \boldsymbol{\mu}_d) \sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

where $\bar{\mathbf{d}} = \frac{1}{n} \sum_{i=1}^n \mathbf{d}_i$ and $\mathbf{S}_d = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{d}_i - \bar{\mathbf{d}})(\mathbf{d}_i - \bar{\mathbf{d}})^\top$

We use the same inference procedures as before:

- Reject $H_0 : \boldsymbol{\mu}_d = \boldsymbol{\mu}_0$ if $(\bar{\mathbf{d}} - \boldsymbol{\mu}_0)^\top \mathbf{S}_d^{-1}(\bar{\mathbf{d}} - \boldsymbol{\mu}_0) > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$
- Use same procedures for forming confidence regions/intervals

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More than Two Repeated Measurements

In a **repeated measures design** units participate in $q > 2$ treatments.

- Assuming a *single* response variable at q treatments
- $\mathbf{X} = \{x_{ij}\}$ is the n units $\times$ q treatments data matrix

We can assume that $\mathbf{x}_i \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

- $\boldsymbol{\mu} = (\mu_1, \dots, \mu_q)$ is the mean vector
- $\boldsymbol{\Sigma}$ is the $q \times q$ covariance matrix

Could use Hotelling's T^2 , but we will consider a new parameterization.

Contrast Matrices

We could consider making contrasts of the component means such as

$$\begin{pmatrix} \mu_1 - \mu_2 \\ \mu_1 - \mu_3 \\ \vdots \\ \mu_1 - \mu_q \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 1 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & -1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_q \end{pmatrix} = \mathbf{C}_1 \boldsymbol{\mu}$$

or

$$\begin{pmatrix} \mu_2 - \mu_1 \\ \mu_3 - \mu_2 \\ \vdots \\ \mu_q - \mu_{q-1} \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix} \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_q \end{pmatrix} = \mathbf{C}_2 \boldsymbol{\mu}$$

which allows us to compare select mean differences.

Hotelling's T^2 with Contrast Matrices

Note that a **contrast matrix** is any matrix $\mathbf{C}_j$ that

1. Has linearly independent rows
2. Satisfies $\mathbf{C}_j \mathbf{1}_q = \mathbf{0}$ (i.e., rows sum to 0)

If $\boldsymbol{\mu} = \mu \mathbf{1}_q$ for some $\mu \in \mathbb{R}$, then $\mathbf{C}_j \boldsymbol{\mu} = \mathbf{0}$ for any contrast matrix $\mathbf{C}_j$.

We can use T^2 to test $H_0 : \boldsymbol{\mu} \propto \mathbf{1}_q$ versus $H_1 : \boldsymbol{\mu} \not\propto \mathbf{1}_q$

$$T^2 = n(\mathbf{C}\bar{\mathbf{x}})^\top (\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1} \mathbf{C}\bar{\mathbf{x}} \sim \frac{(n-1)(q-1)}{n-q+1} F_{q-1, n-q+1}(\alpha)$$

where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and $\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$

Simultaneous T^2 CRs and CIs with Contrast Matrices

Given $\mathbf{C}$, a $100(1 - \alpha)\%$ confidence region (CR) for $\mathbf{C}\boldsymbol{\mu}$ is defined as the set of all $\mathbf{C}\boldsymbol{\mu}$ that satisfy

$$n(\mathbf{C}\bar{\mathbf{x}} - \mathbf{C}\boldsymbol{\mu})^\top (\mathbf{CSC}^\top)^{-1} (\mathbf{C}\bar{\mathbf{x}} - \mathbf{C}\boldsymbol{\mu}) \leq \frac{(n-1)(q-1)}{n-q+1} F_{q-1, n-q+1}$$

This implies that a simultaneous $100(1 - \alpha)\%$ confidence interval (CI) for a single contrast $\mathbf{c}^\top \boldsymbol{\mu}$ has the form

$$\mathbf{c}^\top \bar{\mathbf{x}} \pm \sqrt{\frac{\mathbf{c}^\top \mathbf{S} \mathbf{c}}{n}} \sqrt{\frac{(n-1)(q-1)}{n-q+1} F_{q-1, n-q+1}(\alpha)}$$

R Function for T^2 with Contrast Matrices

```

RM.test <- function(X, mu=0, C=NULL)\{
  X <- as.matrix(X)
  n <- nrow(X)
  p <- ncol(X)
  df2 <- n - p + 1
  if(df2 < 1L) stop("Need nrow(X) > ncol(X).")
  if(length(mu) != p) mu <- rep(mu[1], p)
  xbar <- colMeans(X)
  S <- cov(X)
  if(is.null(C))\{
    C <- matrix(0, p-1, p)
    for(k in 1:(p-1)) C[k, 1:2 + 1*(k-1)] <- c(1, -1)
  \} else \{
    if(nrow(C) != (p-1)) stop("Need [ncol(X)-1] == nrow(C).")
    if(ncol(C) != p) stop("Need ncol(X) == ncol(C).")
    if(any(rowSums(C)>0L)) stop("Need rowSums(C) == rep(0, nrow(C)).")
  \}
  T2 <- n * t(C %*% (xbar - mu)) %*% solve(C %*% S %*% t(C)) %*% (C %*% (xbar - mu))
  Fstat <- T2 / ((p-1) * (n-1) / df2)
  pval <- 1 - pf(Fstat, df1=p-1, df2=df2)
  data.frame(T2=as.numeric(T2), Fstat=as.numeric(Fstat),
             df1=p-1, df2=df2, p.value=as.numeric(pval), row.names="")
\}

```

Example of T^2 with Contrast Matrices in R: H_0 True

```
# RM.test example w/ H0 true (10 data points)
> set.seed(1)
> XX <- matrix(rnorm(10*4), 10, 4)
> RM.test(XX)
      T2      Fstat df1 df2   p.value
1.832424 0.4750728   3   7 0.7094449

# RM.test example w/ H0 true (100 data points)
> set.seed(1)
> XX <- matrix(rnorm(100*4), 100, 4)
> RM.test(XX)
      T2      Fstat df1 df2   p.value
1.286456 0.4201555   3  97 0.7389465

# RM.test example w/ H0 true (500 data points)
> set.seed(1)
> XX <- matrix(rnorm(500*4), 500, 4)
> RM.test(XX)
      T2      Fstat df1 df2   p.value
1.231931 0.4089978   3 497 0.7466049
```


Example of T^2 with Contrast Matrices in R: H_0 False

```
# RM.test example w/ H0 false (10 data points)
> set.seed(1)
> XX <- matrix(rnorm(10*4), 10, 4)
> XX <- XX + matrix(c(0,0,0,0.25), 10, 4, byrow=TRUE)
> RM.test(XX)
      T2      Fstat df1 df2    p.value
2.975373 0.7713929   3   7 0.5456821

# RM.test example w/ H0 false (100 data points)
> set.seed(1)
> XX <- matrix(rnorm(100*4), 100, 4)
> XX <- XX + matrix(c(0,0,0,0.25), 100, 4, byrow=TRUE)
> RM.test(XX)
      T2      Fstat df1 df2    p.value
6.868081 2.243111   3  97 0.08812235

# RM.test example w/ H0 false (500 data points)
> set.seed(1)
> XX <- matrix(rnorm(500*4), 500, 4)
> XX <- XX + matrix(c(0,0,0,0.25), 500, 4, byrow=TRUE)
> RM.test(XX)
      T2      Fstat df1 df2    p.value
19.72918 6.550036   3 497 0.0002380274
```

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Univariate Reminder: Independent Samples t test

To test $H_0 : \mu_1 - \mu_2 = \mu_0$ versus $H_1 : \mu_1 - \mu_2 \neq \mu_0$ we use

$$t = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where

- $\bar{x}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ik}$ is the k -th group's sample mean
- $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ is the pooled variance estimate
- $s_k^2 = \frac{1}{n_k-1} \sum_{i=1}^{n_k} (x_{ik} - \bar{x}_k)^2$ is the k -th group's sample variance

The $100(1 - \alpha)\%$ CI for the difference in population means is

$$(\bar{x}_1 - \bar{x}_2) - t_{n_1+n_2-2}(\alpha/2)s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{x}_1 - \bar{x}_2) + t_{n_1+n_2-2}(\alpha/2)s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Multivariate Independent Samples T^2 Test

Let $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ for $k \in \{1, 2\}$ and assume that the elements of $\{\mathbf{x}_{1i}\}_{i=1}^{n_1}$ and $\{\mathbf{x}_{2i}\}_{i=1}^{n_2}$ are independent of one another.

The pooled estimate of the covariance matrix has the form

$$\mathbf{S}_p = \frac{n_1 - 1}{n_1 + n_2 - 2} \mathbf{S}_1 + \frac{n_2 - 1}{n_1 + n_2 - 2} \mathbf{S}_2$$

where $\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \mathbf{x}_{ki}$ and $\mathbf{S}_k = \frac{1}{n_k - 1} \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top$.

The T^2 test statistic for testing $H_0 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = \boldsymbol{\mu}_0$ has the form

$$T^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - \boldsymbol{\mu}_0)$$

Multivariate Independent Samples T^2 Test (continued)

Assuming that $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ for $k \in \{1, 2\}$, we have that

$$\begin{aligned} T^2 &= (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - [\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2])^\top \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_p \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - [\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2]) \\ &\sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1} \end{aligned}$$

which is an analogue of the one sample T^2 statistic.

This implies that a $100(1 - \alpha)\%$ confidence region can be formed from

$$P \left[T^2 \leq \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}(\alpha) \right] = 1 - \alpha$$

Multivariate Independent Samples T^2 Test R Function

```

T.test <- function(X, Y=NULL, mu=0, paired=FALSE, asymp=FALSE){
  if(is.null(Y)){
    # one-sample T^2 test: same code as before (omitted here)
  } else {
    if(paired){
      # dependent two-sample T^2 test
      X <- as.matrix(X)
      Y <- as.matrix(Y)
      if(!identical(dim(X),dim(Y))) stop("Need dim(X) == dim(Y).")
      xx <- T.test(X-Y, mu=mu, asymp=asymp)
      xx$type <- "dep-sample"
      return(xx)
    } else {
      # independent two-sample T^2 test
      X <- as.matrix(X)
      Y <- as.matrix(Y)
      nx <- nrow(X)
      ny <- nrow(Y)
      p <- ncol(X)
      df2 <- nx + ny - p - 1
      if(p != ncol(Y)) stop("Need ncol(X) == ncol(Y).")
      if(min(nx,ny) <= p) stop("Need min(nrow(X),nrow(Y)) > ncol(X).")
      Sp <- ((nx-1)*cov(X) + (ny-1)*cov(Y)) / (nx + ny - 2)
      dbar <- colMeans(X) - colMeans(Y)
      T2 <- (1/((1/nx) + (1/ny))) * t(dbar - mu) %*% solve(Sp) %*% (dbar - mu)
      Fstat <- T2 / ((nx + ny - 2) * p / df2)
      if(asymp){
        pval <- 1 - pchisq(T2, df=p)
      } else {
        pval <- 1 - pf(Fstat, df1=p, df2=df2)
      }
      return(data.frame(T2=as.numeric(T2), Fstat=as.numeric(Fstat),
                        df1=p, df2=df2, p.value=as.numeric(pval),
                        type="ind-sample", asymp=asymp, row.names=""))
    }
  }
}

```

Multivariate Independent Samples T^2 Test in R

```

> X4 <- subset(mtcars, cyl==4)[,c("mpg","disp","hp","wt")]
> X6 <- subset(mtcars, cyl==6)[,c("mpg","disp","hp","wt")]
> X8 <- subset(mtcars, cyl==8)[,c("mpg","disp","hp","wt")]
> T.test(X4, X6)
      T2      Fstat df1 df2      p.value      type asymp
39.60993 8.045767   4  13 0.001713109 ind-sample FALSE
> T.test(X4, X8)
      T2      Fstat df1 df2      p.value      type asymp
185.6083 40.34963   4  20 2.627159e-09 ind-sample FALSE
> T.test(X6, X8)
      T2      Fstat df1 df2      p.value      type asymp
61.97388 13.04713   4  16 6.545202e-05 ind-sample FALSE

```

Comparing Finite Sample and Large Sample p-values

```
# compare finite sample and large sample p-values (n=10)
> set.seed(1)
> n <- 10
> XX <- matrix(rnorm(n*4), n, 4)
> YY <- matrix(rnorm(n*4), n, 4)
> T.test(XX,YY)
      T2      Fstat df1 df2   p.value      type asymp
3.582189 0.7462893   4  15 0.5754577 ind-sample FALSE
> T.test(XX,YY,asympt=T)
      T2      Fstat df1 df2   p.value      type asymp
3.582189 0.7462893   4  15 0.4654919 ind-sample  TRUE

# compare finite sample and large sample p-values (n=50)
> set.seed(1)
> n <- 50
> XX <- matrix(rnorm(n*4), n, 4)
> YY <- matrix(rnorm(n*4), n, 4)
> T.test(XX,YY)
      T2      Fstat df1 df2   p.value      type asymp
3.286587 0.7964942   4  95 0.530368 ind-sample FALSE
> T.test(XX,YY,asympt=T)
      T2      Fstat df1 df2   p.value      type asymp
3.286587 0.7964942   4  95 0.5110603 ind-sample  TRUE

# compare finite sample and large sample p-values (n=100)
> set.seed(1)
> n <- 100
> XX <- matrix(rnorm(n*4), n, 4)
> YY <- matrix(rnorm(n*4), n, 4)
> T.test(XX,YY)
      T2      Fstat df1 df2   p.value      type asymp
3.270955 0.8053489   4 195 0.5230921 ind-sample FALSE
> T.test(XX,YY,asympt=T)
      T2      Fstat df1 df2   p.value      type asymp
3.270955 0.8053489   4 195 0.5135473 ind-sample  TRUE
```


Simultaneous CIs for Independent Samples T^2

A $100(1 - \alpha)\%$ confidence interval of the form

$$\mathbf{a}^\top (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \pm \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) \mathbf{a}^\top \mathbf{S}_p \mathbf{a}} \sqrt{\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}(\alpha)}$$

covers $\mathbf{a}^\top (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ with probability $1 - \alpha$ simultaneously for all $\mathbf{a}$.

With $\mathbf{a} = \mathbf{e}_j$ (the j -th standard basis vector), we have

$$(\bar{x}_{1j} - \bar{x}_{2j}) \pm \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) s_{jj(p)}} \sqrt{\frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1}(\alpha)}$$

where $s_{jj(p)}$ is the j -th diagonal element of $\mathbf{S}_p$.

Simultaneous CIs for Independent Samples T^2 in R

```
T.ci <- function(mu, Sigma, n, avec=rep(1,length(mu)), level=0.95)\{
  p <- length(mu)
  if(nrow(Sigma)!=p) stop("Need length(mu) == nrow(Sigma).")
  if(ncol(Sigma)!=p) stop("Need length(mu) == ncol(Sigma).")
  if(length(avec)!=p) stop("Need length(mu) == length(avec).")
  if(level <=0 | level >= 1) stop("Need 0 < level < 1.")
  zhat <- crossprod(avec, mu)
  if(length(n)==1L)\{
    cval <- qf(level, p, n-p) * p * (n-1) / (n-p)
    zvar <- crossprod(avec, Sigma %*% avec) / n
  \} else \{
    df2 <- n[1] + n[2] - p - 1
    cval <- qf(level, p, df2) * p * (n[1]+n[2]-2) / df2
    zvar <- crossprod(avec, Sigma %*% avec) * ( (1/n[1]) + (1/n[2]) )
  \}
  const <- sqrt(cval * zvar)
  c(lower = zhat - const, upper = zhat + const)
\}
```

Example of Simultaneous CIs for Indep. Samples T^2

```

> X4 <- subset(mtcars, cyl==4)[,c("mpg","disp","hp","wt")]
> X6 <- subset(mtcars, cyl==6)[,c("mpg","disp","hp","wt")]
> n4 <- nrow(X4)
> n6 <- nrow(X6)
> dbar <- colMeans(X4) - colMeans(X6)
> Sp <- ((n4-1)*cov(X4) + (n6-1)*cov(X6)) / (n4 + n6 - 2)
> dbar
      mpg      disp      hp      wt
6.9207792 -78.1779221 -39.6493506 -0.8314156
> T.ci(dbar, Sp, c(n4,n6), c(1,0,0,0))
      lower      upper
-0.1082001 13.9497585
> T.ci(dbar, Sp, c(n4,n6), c(0,1,0,0))
      lower      upper
-141.59106 -14.76478
> T.ci(dbar, Sp, c(n4,n6), c(0,0,1,0))
      lower      upper
-82.189575  2.890874
> T.ci(dbar, Sp, c(n4,n6), c(0,0,0,1))
      lower      upper
-1.7885075  0.1256764

```

T^2 Test with Heterogeneous Covariances ($\Sigma_1 \neq \Sigma_2$)

Let $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \Sigma_k)$ for $k \in \{1, 2\}$ and assume that the elements of $\{\mathbf{x}_{1i}\}_{i=1}^{n_1}$ and $\{\mathbf{x}_{2i}\}_{i=1}^{n_2}$ are independent of one another.

We cannot define a “distance” measure like T^2 , whose distribution does not depend on the unknown population parameters Σ_1 and Σ_2 .

Use the modified T^2 statistic with non-pooled covariance matrices

$$T^2 = (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - [\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2])^\top [n_1^{-1} \mathbf{S}_1 + n_2^{-1} \mathbf{S}_2]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - [\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2])$$

where $\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \mathbf{x}_{ki}$ and $\mathbf{S}_k = \frac{1}{n_k - 1} \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top$.

Large and Small Sample T^2 Inferences when $\Sigma_1 \neq \Sigma_2$

If $\min(n_1, n_2) - p$ is large, we can use the large sample approximation:

$$P[T^2 \leq \chi_p^2(\alpha)] \approx 1 - \alpha$$

which (asymptotically) works for non-normal multivariate data too!

If $\min(n_1, n_2) - p$ is small and we assume normality, we can use

$$T^2 \approx \frac{\nu p}{\nu - p + 1} F_{p, \nu - p + 1}$$

where the degrees of freedom parameter ν is estimated as

$$\nu = \frac{p + p^2}{\sum_{k=1}^2 \frac{1}{n_k} \left\{ \text{tr} \left[\left(\frac{1}{n_k} \mathbf{S}_k \mathbf{S}_0^{-1} \right)^2 \right] + \left(\text{tr} \left[\frac{1}{n_k} \mathbf{S}_k \mathbf{S}_0^{-1} \right] \right)^2 \right\}}$$

with $\mathbf{S}_0 = \frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2$. Note that $\min(n_1, n_2) \leq \nu \leq n_1 + n_2$.

Add var.equal Option to T^2 Test R Function

```
T.test <- function(X, Y=NULL, mu=0, paired=FALSE, asymp=FALSE, var.equal=TRUE)\{
  if(is.null(Y))\{
    # one-sample T^2 test: same code as before (omitted here)
  \} else \{
    if(paired)\{
      # dependent two-sample T^2 test: same code as before (omitted here)
    \} else \{
      # independent two-sample T^2 test
      X <- as.matrix(X)
      Y <- as.matrix(Y)
      nx <- nrow(X)
      ny <- nrow(Y)
      p <- ncol(X)
      if(p != ncol(Y)) stop("Need ncol(X) == ncol(Y).")
      if(min(nx,ny) <= p) stop("Need min(nrow(X),nrow(Y)) > ncol(X).")
      dbar <- colMeans(X) - colMeans(Y)
      if(var.equal)\{
        df2 <- nx + ny - p - 1
        Sp <- ((nx-1)*cov(X) + (ny-1)*cov(Y)) / (nx + ny - 2)
        T2 <- (1/((1/nx) + (1/ny))) * t(dbar - mu) %*% solve(Sp) %*% (dbar - mu)
        Fstat <- T2 / ((nx + ny - 2) * p / df2)
      \} else \{
        Sx <- cov(X)
        Sy <- cov(Y)
        Sp <- (Sx/nx) + (Sy/ny)
        T2 <- t(dbar - mu) %*% solve(Sp) %*% (dbar - mu)
        SpInv <- solve(Sp)
        SxSpInv <- (1/nx) * Sx %*% SpInv
        SySpInv <- (1/ny) * Sy %*% SpInv
        nudx <- (sum(diag(SxSpInv %*% SxSpInv)) + (sum(diag(SxSpInv)))^2) / nx
        nudy <- (sum(diag(SySpInv %*% SySpInv)) + (sum(diag(SySpInv)))^2) / ny
        nu <- (p + p^2) / (nudx + nudy)
        df2 <- nu - p + 1
        Fstat <- T2 / (nu * p / df2)
      \}
      if(asymp)\{
        pval <- 1 - pchisq(T2, df=p)
      \} else \{
        pval <- 1 - pf(Fstat, df1=p, df2=df2)
      \}
      return(data.frame(T2=as.numeric(T2), Fstat=as.numeric(Fstat),
        df1=p, df2=df2, p.value=as.numeric(pval),
        type="ind-sample", asymp=asymp, var.equal=var.equal, row.names=""))
    }
  }
```

Example with var.equal Option

```
> X4 <- subset(mtcars, cyl==4)[,c("mpg","disp","hp","wt")]
> X6 <- subset(mtcars, cyl==6)[,c("mpg","disp","hp","wt")]
> T.test(X4, X6)
      T2      Fstat df1 df2      p.value      type asymp var.equal
39.60993 8.045767   4  13 0.001713109 ind-sample FALSE      TRUE
> T.test(X4, X6, var.equal=FALSE)
      T2      Fstat df1      df2      p.value      type asymp var.equal
46.04706 9.266334   4 12.38026 0.001067989 ind-sample FALSE      FALSE

> set.seed(1)
> n <- 100
> XX <- matrix(rnorm(n*4), n, 4)
> YY <- matrix(rnorm(n*4), n, 4)
> T.test(XX,YY)
      T2      Fstat df1 df2      p.value      type asymp var.equal
3.270955 0.8053489   4 195 0.5230921 ind-sample FALSE      TRUE
> T.test(XX,YY,var.equal=F)
      T2      Fstat df1      df2      p.value      type asymp var.equal
3.270955 0.8053198   4 194.537 0.5231144 ind-sample FALSE      FALSE
```

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Univariate Reminder: One-Way ANOVA

Suppose that $x_{ki} \stackrel{\text{iid}}{\sim} N(\mu_k, \sigma^2)$ for $k \in \{1, \dots, g\}$ and $i \in \{1, \dots, n_k\}$.

The one-way analysis of variance (ANOVA) model has the form

$$x_{ki} = \mu + \alpha_k + \epsilon_{ki}$$

where μ is the overall mean, α_k is the k -th group's treatment effect with the constraint that $\sum_{k=1}^g n_k \alpha_k = 0$, and $\epsilon_{ki} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ are error terms.

The sample estimates of the model parameters are

$$\hat{\mu} = \bar{x} \quad \text{and} \quad \hat{\alpha}_k = \bar{x}_k - \bar{x} \quad \text{and} \quad \hat{\epsilon}_{ki} = x_{ki} - \bar{x}_k$$

where $\bar{x} = \frac{1}{\sum_{k=1}^g n_k} \sum_{k=1}^g n_k \bar{x}_k$ and $\bar{x}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{ki}$.

Univariate Reminder: One-Way ANOVA (continued)

We want to test the hypotheses $H_0 : \alpha_k = 0$ for all $k \in \{1, \dots, g\}$ versus $H_1 : \alpha_k \neq 0$ for some $k \in \{1, \dots, g\}$.

The decomposition of the sums-of-squares has the form

$$\underbrace{\sum_{k=1}^g \sum_{i=1}^{n_k} (x_{ki} - \bar{x})^2}_{\text{SS Total}} = \underbrace{\sum_{k=1}^g n_k (\bar{x}_k - \bar{x})^2}_{\text{SS Between}} + \underbrace{\sum_{k=1}^g \sum_{i=1}^{n_k} (x_{ki} - \bar{x}_k)^2}_{\text{SS Within}}$$

The ANOVA F test rejects H_0 at level α if

$$F = \frac{SSB/(g-1)}{SSW/(n-g)} > F_{g-1, n-g}(\alpha)$$

where $n = \sum_{k=1}^g n_k$.

Multivariate Extension of One-Way ANOVA

Let $\mathbf{x}_{ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$ for $k \in \{1, \dots, g\}$ and assume that the elements of $\{\mathbf{x}_{ki}\}_{i=1}^{n_k}$ and $\{\mathbf{x}_{\ell i}\}_{i=1}^{n_\ell}$ are independent of one another.

The one-way multivariate analysis of variance (MANOVA) has the form

$$\mathbf{x}_{ki} = \boldsymbol{\mu} + \boldsymbol{\alpha}_k + \boldsymbol{\epsilon}_{ki}$$

where $\boldsymbol{\mu}_k = \boldsymbol{\mu} + \boldsymbol{\alpha}_k$, $\boldsymbol{\mu}$ is the overall mean vector, $\boldsymbol{\alpha}_k$ is the k -th group's treatment effect vector (with $\sum_{k=1}^g n_k \boldsymbol{\alpha}_k = \mathbf{0}_p$), and $\boldsymbol{\epsilon}_{ki} \stackrel{\text{iid}}{\sim} N(\mathbf{0}_p, \boldsymbol{\Sigma})$.

The sample estimates of the model parameters are

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}} \quad \text{and} \quad \hat{\boldsymbol{\alpha}}_k = \bar{\mathbf{x}}_k - \bar{\mathbf{x}} \quad \text{and} \quad \hat{\boldsymbol{\epsilon}}_{ki} = \mathbf{x}_{ki} - \bar{\mathbf{x}}_k$$

where $\bar{\mathbf{x}} = \frac{1}{\sum_{k=1}^g n_k} \sum_{k=1}^g n_k \bar{\mathbf{x}}_k$ and $\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \mathbf{x}_{ki}$.

Sums-of-Squares and Crossproducts Decomposition

The MANOVA sums-of-squares and crossproducts decomposition is

$$\underbrace{\sum_{k=1}^g \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}})(\mathbf{x}_{ki} - \bar{\mathbf{x}})^\top}_{\text{SSCP Total}} = \underbrace{\sum_{k=1}^g n_k (\bar{\mathbf{x}}_k - \bar{\mathbf{x}})(\bar{\mathbf{x}}_k - \bar{\mathbf{x}})^\top}_{\text{SSCP Between}} + \underbrace{\sum_{k=1}^g \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top}_{\text{SSCP Within}}$$

and note that the within SSCP matrix has the form

$$\begin{aligned} \mathbf{W} &= \sum_{k=1}^g \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top \\ &= \sum_{k=1}^g (n_k - 1) \mathbf{S}_k \end{aligned}$$

where $\mathbf{S}_k = \frac{1}{n_k - 1} \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top$ is the k -th group's sample covariance matrix.

One-Way MANOVA SSCP Table

Similar to the one-way ANOVA, we can summarize the SSCP information in a table

Source	SSCP Matrix	D.F.
Between	$\mathbf{B} = \sum_{k=1}^g n_k (\bar{\mathbf{x}}_k - \bar{\mathbf{x}})(\bar{\mathbf{x}}_k - \bar{\mathbf{x}})^\top$	$g - 1$
Within	$\mathbf{W} = \sum_{k=1}^g \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)(\mathbf{x}_{ki} - \bar{\mathbf{x}}_k)^\top$	$\sum_{k=1}^g n_k - g$
Total	$\mathbf{B} + \mathbf{W} = \sum_{k=1}^g \sum_{i=1}^{n_k} (\mathbf{x}_{ki} - \bar{\mathbf{x}})(\mathbf{x}_{ki} - \bar{\mathbf{x}})^\top$	$\sum_{k=1}^g n_k - 1$

Between = treatment sum-of-squares and crossproducts

Within = residual (error) sum-of-squares and crossproducts

MANOVA Extension of ANOVA F Test

We want to test the hypotheses $H_0 : \boldsymbol{\alpha}_k = \mathbf{0}_p$ for all $k \in \{1, \dots, g\}$ versus $H_1 : \boldsymbol{\alpha}_k \neq \mathbf{0}_p$ for some $k \in \{1, \dots, g\}$.

The MANOVA test statistic has the form

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|}$$

which is known as **Wilks' lambda**.

Reject H_0 if Λ^* is smaller than expected under H_0 .

Distribution for Wilks' Lambda

For certain special cases, the exact distribution of Λ^* is known

p	g	Sampling Distribution
$p = 1$	$g \geq 2$	$\left(\frac{n-g}{g-1}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{g-1, n-g}$
$p = 2$	$g \geq 2$	$\left(\frac{n-g-1}{g-1}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2(g-1), 2(n-g-1)}$
$p \geq 1$	$g = 2$	$\left(\frac{n-p-1}{p}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim F_{p, n-p-1}$
$p \geq 1$	$g = 3$	$\left(\frac{n-p-2}{p}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim F_{2p, 2(n-p-2)}$

where $n = \sum_{k=1}^g n_k$.

If n is large and H_0 is true, then

$$-\left(n - 1 - \frac{p+g}{2}\right) \log(\Lambda^*) \approx \chi_{p(g-1)}^2$$

Other One-Way MANOVA Test Statistics

There are other popular MANOVA test statistics

- Lawley-Hotelling trace: $\text{tr}(\mathbf{B}\mathbf{W}^{-1})$
- Pillai trace: $\text{tr}(\mathbf{B}[\mathbf{B} + \mathbf{W}]^{-1})$
- Roy's largest root: maximum eigenvalue of $\mathbf{W}(\mathbf{B} + \mathbf{W})^{-1}$

Each of these test statistics has a corresponding (approximate) distribution, and all should produce similar inference for large n .

- Some evidence that Pillai's trace is more robust to non-normality

One-Way MANOVA Example in R

```
> X <- as.matrix(mtcars[,c("mpg", "disp", "hp", "wt")])
> cylinder <- factor(mtcars$cyl)
> mod <- lm(X ~ cylinder)
> Manova(mod, test.statistic="Pillai")
```

Type II MANOVA Tests: Pillai test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	1.0838	7.9845	8	54	4.969e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> Manova(mod, test.statistic="Wilks")
```

Type II MANOVA Tests: Wilks test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	0.091316	15.01	8	52	4e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> Manova(mod, test.statistic="Roy")
```

Type II MANOVA Tests: Roy test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	7.7873	52.564	4	27	2.348e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> Manova(mod, test.statistic="Hotelling-Lawley")
```

Type II MANOVA Tests: Hotelling-Lawley test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	8.0335	25.105	8	50	5.341e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Variance of Treatment Effect Difference

Let α_{kj} denote the j -th element of $\boldsymbol{\alpha}_k$ and note that

$$\hat{\alpha}_{kj} = \bar{x}_{kj} - \bar{x}_j$$

where $\bar{x}_j = \frac{1}{\sum_{k=1}^g n_k} \sum_{k=1}^g n_k \bar{x}_{kj}$ and $\bar{x}_{kj} = \frac{1}{n_k} \sum_{i=1}^{n_k} x_{kij}$

- $\bar{x}_{kj}$ is the k -th group's mean for the j -th variable
- $\bar{x}_j$ is the overall mean of the j -th variable

The variance for the difference in the estimated treatment effects is

$$\text{Var}(\hat{\alpha}_{kj} - \hat{\alpha}_{\ell j}) = \text{Var}(\bar{x}_{kj} - \bar{x}_{\ell j}) = \left(\frac{1}{n_k} + \frac{1}{n_\ell} \right) \sigma_{jj}$$

where σ_{jj} is the j -th diagonal element of $\boldsymbol{\Sigma}$.

Forming Simultaneous CIs via Bonferroni's Method

To estimate the variance for the treatment effect difference, we use

$$\hat{\text{Var}}(\hat{\alpha}_{kj} - \hat{\alpha}_{\ell j}) = \left(\frac{1}{n_k} + \frac{1}{n_\ell} \right) \frac{w_{jj}}{n - g}$$

where w_{jj} denotes the j -th diagonal of $\mathbf{W}$.

We can use Bonferroni's method to control the familywise error rate

- Have p variables and $g(g - 1)/2$ pairwise comparisons
- Total of $q = pg(g - 1)/2$ tests to control for
- Use critical values $t_{n-g}(\alpha/[2q]) = t_{n-g}(\alpha/[pg(g - 1)])$

Get Least-Squares MANOVA Means in R

```
# get least-squares means for each variable
> library(emmeans)
> p <- ncol(X)
> lsm <- vector("list", p)
> names(lsm) <- colnames(X)
> for(j in 1:p){
+   wts <- rep(0, p)
+   wts[j] <- 1
+   lsm[[j]] <- emmeans(mod, "cylinder", weights=wts)
+ }
```

```
> lsm[[1]]
cylinder emmean    SE df lower.CL upper.CL
4          26.7 0.972 29    24.7    28.7
6          19.7 1.218 29    17.3    22.2
8          15.1 0.861 29    13.3    16.9
```

Results are averaged over the levels of: rep.meas
Confidence level used: 0.95

```
> lsm[[3]]
cylinder emmean    SE df lower.CL upper.CL
4          82.6 11.4 29    59.3    106
6         122.3 14.3 29    93.0    152
8         209.2 10.1 29   188.5    230
```

Results are averaged over the levels of: rep.meas
Confidence level used: 0.95

Form Simultaneous Confidence Intervals in R

```
# get alpha level for Bonferroni correction
> q <- p * 3 * (3-1) / 2
> alpha <- 0.05 / q

# Bonferroni pairwise CIs for "mpg"
> confint(contrast(lsm[[1]], "pairwise"), level=1-alpha, adj="none")
  contrast      estimate    SE df lower.CL upper.CL
cylinder4 - cylinder6      6.92 1.56 29   2.07362    11.77
cylinder4 - cylinder8     11.56 1.30 29   7.52433    15.60
cylinder6 - cylinder8      4.64 1.49 29   0.00205     9.28
```

Results are averaged over the levels of: rep.meas
Confidence level used: 0.995833333333333

```
# Bonferroni pairwise CIs for "hp"
> confint(contrast(lsm[[3]], "pairwise"), level=1-alpha, adj="none")
  contrast      estimate    SE df lower.CL upper.CL
cylinder4 - cylinder6    -39.6 18.3 29   -96.7     17.4
cylinder4 - cylinder8   -126.6 15.3 29  -174.1    -79.1
cylinder6 - cylinder8    -86.9 17.6 29  -141.5    -32.3
```

Results are averaged over the levels of: rep.meas
Confidence level used: 0.995833333333333

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Testing the Homogeneity of Covariances Assumption

The one-way MANOVA model assumes that $\Sigma_1 = \cdots = \Sigma_g$.

To test $H_0 : \Sigma_1 = \cdots = \Sigma_g$ versus $H_1 : \Sigma_k \neq \Sigma_\ell$ for some $k, \ell \in \{1, \dots, g\}$, we use the likelihood ratio test (LRT) statistic

$$\Lambda = \prod_{k=1}^g \left(\frac{|\mathbf{S}_k|}{|\mathbf{S}_P|} \right)^{(n_k-1)/2}$$

where $\mathbf{S}_P = \frac{1}{n-g} \mathbf{W}$ is the pooled covariance matrix estimate.

It was shown (by George Box) that $-2 \log(\Lambda) \approx \frac{1}{1-u} \chi_\nu^2$ where

$$u = \left(\sum_{k=1}^g \frac{1}{n_k} - \frac{1}{n} \right) \left(\frac{2p^2 + 3p - 1}{6(p+1)(g-1)} \right)$$

$$\nu = p(p+1)(g-1)/2$$

Testing Homogeneity of Covariances in R

```
> library(biotools)
> boxM(X, cylinder)
```

Box's M-test for Homogeneity of Covariance Matrices

```
data: X
Chi-Sq (approx.) = 40.851, df = 20, p-value = 0.003892
```

We reject the null hypothesis that $\Sigma_1 = \cdots = \Sigma_g$, so our previous MANOVA results may be invalid.

M -test is sensitive to non-normality, so it may be ok to proceed with the MANOVA in some cases when the M -test rejects H_0 .

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Univariate Reminder: Two-Way ANOVA

$x_{\ell ki} \stackrel{\text{iid}}{\sim} N(\mu_{k\ell}, \sigma^2)$ for $k \in \{1, \dots, a\}$, $\ell \in \{1, \dots, b\}$, and $i \in \{1, \dots, n\}$.

The two-way analysis of variance (ANOVA) model has the form

$$x_{\ell ki} = \mu + \alpha_k + \beta_\ell + \gamma_{k\ell} + \epsilon_{\ell ki}$$

where

- μ is the overall mean
- α_k is the main effect for factor 1 ($\sum_{k=1}^a \alpha_k = 0$)
- β_ℓ is the main effect for factor 2 ($\sum_{\ell=1}^b \beta_\ell = 0$)
- $\gamma_{k\ell}$ is the interaction effect between factors 1 and 2 ($\sum_{k=1}^a \gamma_{k\ell} = \sum_{\ell=1}^b \gamma_{k\ell} = 0$)
- $\epsilon_{\ell ki} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ are error terms

Univariate Reminder: Two-Way ANOVA Estimation

The two-way ANOVA model implies the decomposition

$$x_{\ell ki} = \underbrace{\bar{x}}_{\hat{\mu}} + \underbrace{(\bar{x}_{k\cdot} - \bar{x})}_{\hat{\alpha}_k} + \underbrace{(\bar{x}_{\cdot\ell} - \bar{x})}_{\hat{\beta}_\ell} + \underbrace{(\bar{x}_{k\ell} - \bar{x}_{k\cdot} - \bar{x}_{\cdot\ell} + \bar{x})}_{\hat{\gamma}_{k\ell}} + \underbrace{(x_{\ell ki} - \bar{x}_{k\ell})}_{\hat{\epsilon}_{\ell ki}}$$

where

- $\bar{x} = \frac{1}{abn} \sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n x_{\ell ki}$ is the overall mean
- $\bar{x}_{k\cdot} = \frac{1}{bn} \sum_{\ell=1}^b \sum_{i=1}^n x_{\ell ki}$ is the mean of k -th level of factor 1
- $\bar{x}_{\cdot\ell} = \frac{1}{an} \sum_{k=1}^a \sum_{i=1}^n x_{\ell ki}$ is the mean of ℓ -th level of factor 2
- $\bar{x}_{k\ell} = \frac{1}{n} \sum_{i=1}^n x_{\ell ki}$ is the mean of the k -th level of factor 1 and the ℓ -th level of factor 2

Univariate Reminder: Two-Way ANOVA Sum-of-Sq

The two-way ANOVA sum-of-squares decomposition is

$$\begin{aligned}
 \underbrace{\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (x_{\ell ki} - \bar{x})^2}_{SSTotal} &= \underbrace{\sum_{k=1}^a bn(\bar{x}_{k\cdot} - \bar{x})^2}_{SSA} + \underbrace{\sum_{\ell=1}^b an(\bar{x}_{\cdot\ell} - \bar{x})^2}_{SSB} \\
 &+ \underbrace{\sum_{k=1}^a \sum_{\ell=1}^b n(\bar{x}_{k\ell} - \bar{x}_{k\cdot} - \bar{x}_{\cdot\ell} + \bar{x})^2}_{SSAB} \\
 &+ \underbrace{\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (x_{\ell ki} - \bar{x}_{k\ell})^2}_{SSError}
 \end{aligned}$$

Univariate Reminder: Two-Way ANOVA Inference

Source	Sum-of-Squares	D.F.
Factor 1	$\sum_{k=1}^a bn(\bar{x}_{k\cdot} - \bar{x})^2$	$a - 1$
Factor 2	$\sum_{\ell=1}^b an(\bar{x}_{\cdot\ell} - \bar{x})^2$	$b - 1$
Interaction	$\sum_{k=1}^a \sum_{\ell=1}^b n(\bar{x}_{k\ell} - \bar{x}_{k\cdot} - \bar{x}_{\cdot\ell} + \bar{x})^2$	$(a - 1)(b - 1)$
Error	$\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (x_{\ell ki} - \bar{x}_{k\ell})^2$	$ab(n - 1)$
Total	$\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (x_{\ell ki} - \bar{x})^2$	$abn - 1$

Reject $H_0 : \alpha_1 = \cdots = \alpha_a = 0$ if $\frac{SSA/(a-1)}{SSE/[ab(n-1)]} > F_{a-1, ab(n-1)}(\alpha)$

Reject $H_0 : \beta_1 = \cdots = \beta_b = 0$ if $\frac{SSB/(b-1)}{SSE/[ab(n-1)]} > F_{b-1, ab(n-1)}(\alpha)$

Reject $H_0 : \gamma_{11} = \cdots = \gamma_{k\ell} = 0$ if $\frac{SSAB/[(a-1)(b-1)]}{SSE/[ab(n-1)]} > F_{(a-1)(b-1), ab(n-1)}(\alpha)$

Multivariate Extension of Two-Way ANOVA

$\mathbf{x}_{\ell ki} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_{k\ell}, \boldsymbol{\Sigma})$ for $k \in \{1, \dots, a\}$, $\ell \in \{1, \dots, b\}$, and $i \in \{1, \dots, n\}$.

The two-way multivariate analysis of variance (MANOVA) has the form

$$\mathbf{x}_{\ell ki} = \boldsymbol{\mu} + \boldsymbol{\alpha}_k + \boldsymbol{\beta}_\ell + \boldsymbol{\gamma}_{k\ell} + \boldsymbol{\epsilon}_{\ell ki}$$

where the terms are analogues of those in the two-way ANOVA model.

The sample estimates of the model parameters are

$$\mathbf{x}_{\ell ki} = \underbrace{\bar{\mathbf{x}}}_{\hat{\boldsymbol{\mu}}} + \underbrace{(\bar{\mathbf{x}}_{k\cdot} - \bar{\mathbf{x}})}_{\hat{\boldsymbol{\alpha}}_k} + \underbrace{(\bar{\mathbf{x}}_{\cdot\ell} - \bar{\mathbf{x}})}_{\hat{\boldsymbol{\beta}}_\ell} + \underbrace{(\bar{\mathbf{x}}_{k\ell} - \bar{\mathbf{x}}_{k\cdot} - \bar{\mathbf{x}}_{\cdot\ell} + \bar{\mathbf{x}})}_{\hat{\boldsymbol{\gamma}}_{k\ell}} + \underbrace{(\mathbf{x}_{\ell ki} - \bar{\mathbf{x}}_{k\ell})}_{\hat{\boldsymbol{\epsilon}}_{\ell ki}}$$

where the terms are analogues of those in the two-way ANOVA model.

Two-Way MANOVA Table

Source	SSCP	D.F.
Factor 1	$\sum_{k=1}^a bn(\bar{\mathbf{x}}_{k\cdot} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{k\cdot} - \bar{\mathbf{x}})^\top$	$a - 1$
Factor 2	$\sum_{\ell=1}^b an(\bar{\mathbf{x}}_{\cdot\ell} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{\cdot\ell} - \bar{\mathbf{x}})^\top$	$b - 1$
Interaction	$\sum_{k=1}^a \sum_{\ell=1}^b n\tilde{\mathbf{x}}_{k\ell}\tilde{\mathbf{x}}_{k\ell}^\top$	$(a - 1)(b - 1)$
Error	$\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (\mathbf{x}_{\ell ki} - \bar{\mathbf{x}}_{k\ell})(\mathbf{x}_{\ell ki} - \bar{\mathbf{x}}_{k\ell})^\top$	$ab(n - 1)$
Total	$\sum_{k=1}^a \sum_{\ell=1}^b \sum_{i=1}^n (\mathbf{x}_{\ell ki} - \bar{\mathbf{x}})(\mathbf{x}_{\ell ki} - \bar{\mathbf{x}})^\top$	$abn - 1$

where $\tilde{\mathbf{x}}_{k\ell} = (\bar{\mathbf{x}}_{k\ell} - \bar{\mathbf{x}}_{k\cdot} - \bar{\mathbf{x}}_{\cdot\ell} + \bar{\mathbf{x}})$

$$\text{SSCP}_T = \text{SSCP}_A + \text{SSCP}_B + \text{SSCP}_{AB} + \text{SSCP}_E$$

Two-Way MANOVA Inference

Reject $H_0 : \alpha_1 = \cdots = \alpha_a = \mathbf{0}_p$ if $\nu_A \log(\Lambda_A^*) > \chi_{(a-1)p}^2(\alpha)$

- $\Lambda_A^* = \frac{|\text{SSCP}_E|}{|\text{SSCP}_A + \text{SSCP}_E|}$ and $\nu_A = - \left[ab(n-1) - \frac{p+1-(a-1)}{2} \right]$

Reject $H_0 : \beta_1 = \cdots = \beta_b = \mathbf{0}_p$ if $\nu_B \log(\Lambda_B^*) > \chi_{(b-1)p}^2(\alpha)$

- $\Lambda_B^* = \frac{|\text{SSCP}_E|}{|\text{SSCP}_B + \text{SSCP}_E|}$ and $\nu_B = - \left[ab(n-1) - \frac{p+1-(b-1)}{2} \right]$

Reject $H_0 : \gamma_{11} = \cdots = \gamma_{kl} = \mathbf{0}_p$ if $\nu_{AB} \log(\Lambda_{AB}^*) > \chi_{(a-1)(b-1)p}^2(\alpha)$

- $\Lambda_{AB}^* = \frac{|\text{SSCP}_E|}{|\text{SSCP}_{AB} + \text{SSCP}_E|}$ and $\nu_{AB} = - \left[ab(n-1) - \frac{p+1-(a-1)(b-1)}{2} \right]$

Use $t_{ab(n-1)}$ distribution with Bonferroni correction for CIs.

Two-Way MANOVA Example in R (fit model)

```
# two-way manova with interaction
> data(mtcars)
> X <- as.matrix(mtcars[,c("mpg","displ","hp","wt")])
> cylinder <- factor(mtcars$cyl)
> transmission <- factor(mtcars$am)
> mod <- lm(X ~ cylinder * transmission)
> Manova(mod, test.statistic="Wilks")
```

Type II MANOVA Tests: Wilks test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	0.09550	12.8570	8	46	1.689e-09 ***
transmission	1	0.43720	7.4019	4	23	0.0005512 ***
cylinder:transmission	2	0.58187	1.7880	8	46	0.1040156

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# refit additive model
> mod <- lm(X ~ cylinder + transmission)
> Manova(mod, test.statistic="Wilks")
```

Type II MANOVA Tests: Wilks test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
cylinder	2	0.11779	11.9610	8	50	2.414e-09 ***
transmission	1	0.49878	6.2805	4	25	0.001217 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Two-Way MANOVA Example in R (get LS means)

```
> p <- ncol(X)
> lsm.cyl <- lsm.trn <- vector("list", p)
> names(lsm) <- colnames(X)
> for(j in 1:p){
+   wts <- rep(0, p*2)
+   wts[1:2 + (j-1)*2] <- 1
+   lsm.cyl[[j]] <- emmeans(mod, "cylinder", weights=wts)
+   wts <- rep(0, p*3)
+   wts[1:3 + (j-1)*3] <- 1
+   lsm.trn[[j]] <- emmeans(mod, "transmission", weights=wts)
+ }
```

```
# print mpg LS mean for cylinder effect
> lsm.cyl[[1]]
cylinder emmean    SE df lower.CL upper.CL
4          26.1 0.972 28     24.1     28.1
6          19.9 1.165 28     17.5     22.3
8          16.0 0.943 28     14.1     17.9
```

Results are averaged over the levels of: transmission, rep.meas
Confidence level used: 0.95

```
# print mpg LS mean for transmission effect
> lsm.trn[[1]]
transmission emmean    SE df lower.CL upper.CL
0           19.4 0.797 28     17.8     21.0
1           22.0 0.928 28     20.1     23.9
```

Results are averaged over the levels of: cylinder, rep.meas
Confidence level used: 0.95

Two-Way MANOVA Example in R (simultaneous CIs)

```
# get alpha level for Bonferroni correction
> q <- p * (3 * (3-1) / 2 + 2 * (2-1) / 2)
> alpha <- 0.05 / q

# Bonferroni pairwise CIs for "mpg" (cylinder effect)
> confint(contrast(lsm.cyl[[1]], "pairwise"), level=1-alpha, adj="none")
  contrast      estimate      SE df lower.CL upper.CL
cylinder4 - cylinder6      6.16 1.54 28      1.190     11.12
cylinder4 - cylinder8     10.07 1.45 28      5.372     14.76
cylinder6 - cylinder8      3.91 1.47 28     -0.843      8.67
```

Results are averaged over the levels of: transmission, rep.meas
Confidence level used: 0.996875

```
# Bonferroni pairwise CIs for "mpg" (transmission effect)
> confint(contrast(lsm.trn[[1]], "pairwise"), level=1-alpha, adj="none")
  contrast      estimate      SE df lower.CL upper.CL
transmission0 - transmission1     -2.56 1.3 28     -6.76      1.64
```

Results are averaged over the levels of: cylinder, rep.meas
Confidence level used: 0.996875